AJ Motors

- Working distributed system · implementation and verification
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- https://github.com/AhmadAlkaseb/aj

What works locally?

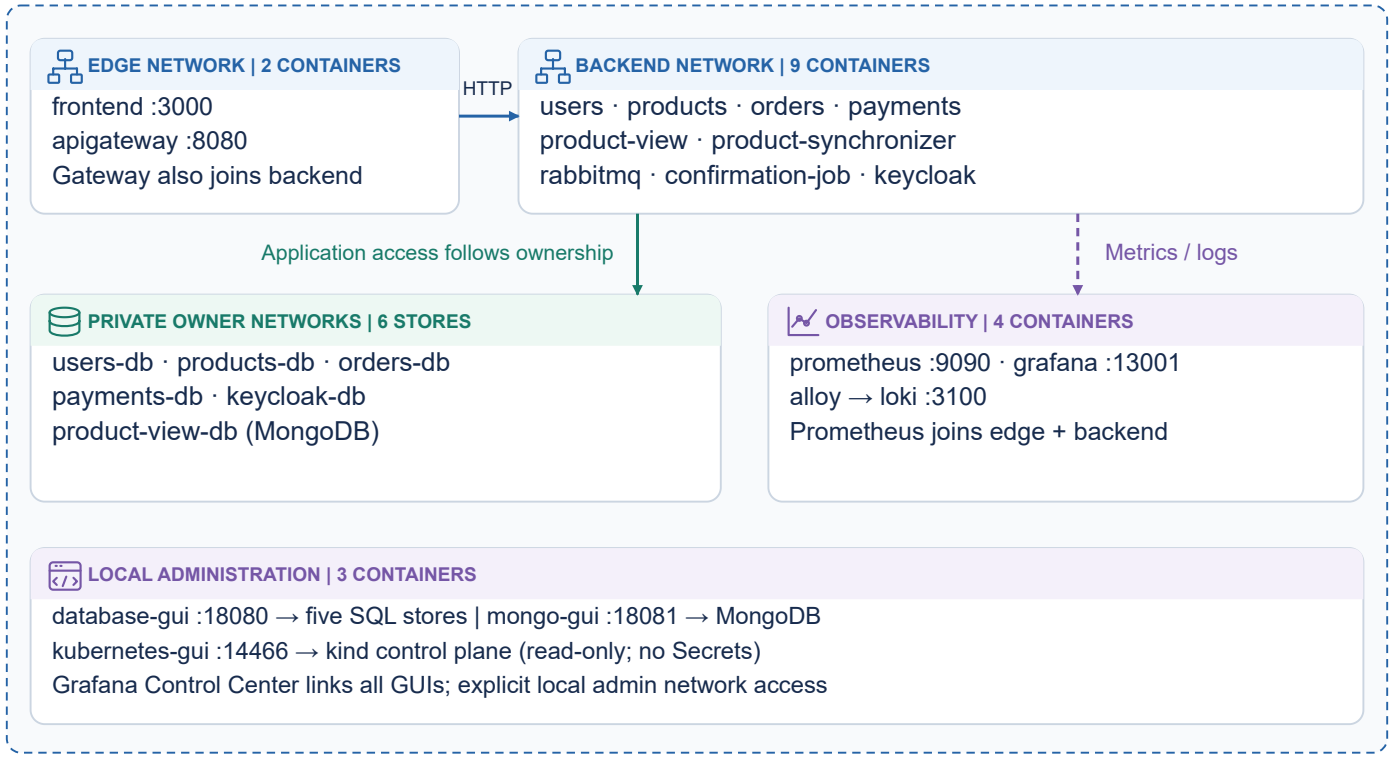
- Catalogue, customer profiles, exclusive reservation and manual payment.
- Real REST/GraphQL frontend workflow and protected backend ownership.
- Actual local AI drafts; bounded PDF confirmation work.
- Selected failure/recovery scenarios and persistent demonstration data.

Compose development architecture

- 24 named services under aj; persistent owner stores and shared local administration.

Docker Compose system architecture

One Docker Desktop project: aj | 24 named containers | loopback host access



Named volumes retain SQL, MongoDB, broker state and monitoring history.

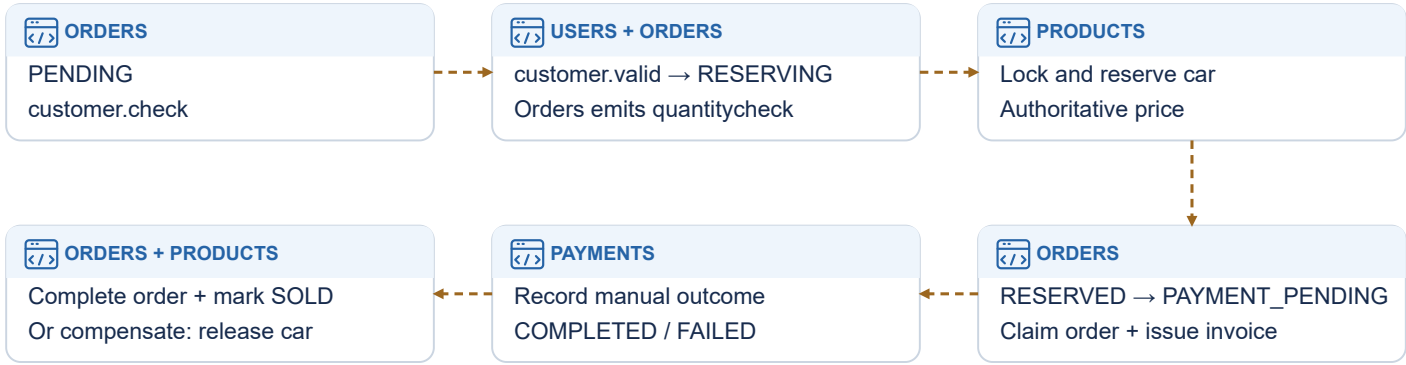
The kind control-plane container is separate from the 24-service Compose project.

Asynchronous cooperation

- Order → customer validation → inventory reservation → payment outcome.
- Failure/cancellation compensates eligible stock.
- Backend cooperation fixture starts private stores, broker and four owners.
- No UI or gateway is required for the dedicated cooperation scenario.

Reservation and payment saga

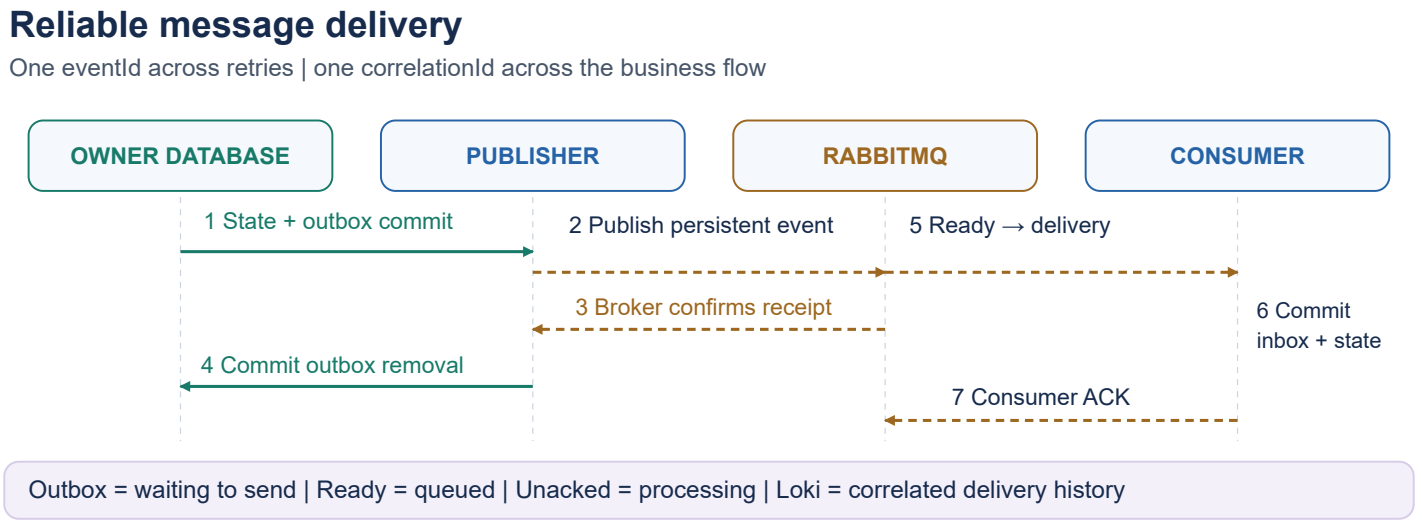
Each arrow crosses a message boundary; each service commits its own local transaction



Simplified successful path. Validation rejection, cancellation and deadlines can branch to recovery.

Delivery and failure visibility

- Outbox: waiting to send; Ready: waiting in broker; Unacked: processing.
- Publisher confirmation differs from consumer acknowledgement.
- Correlated metadata links requests, events and logs without payload secrets.

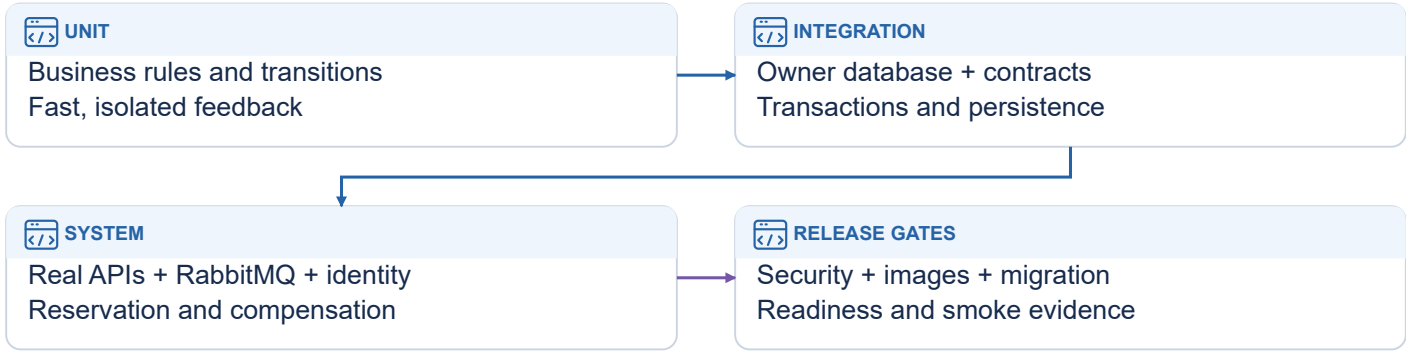


Testing and CI gates

- Unit, API/database/message integration and backend cooperation are separate.
- Change-aware CI skips unrelated work; required failures block dependants.
- 47 local Python policy/deployment/federation tests passed at review.
- GitHub-hosted execution remains unverified without account authentication.

Verification from local rules to business flows

Different test boundaries answer different questions; local evidence does not imply a remote release



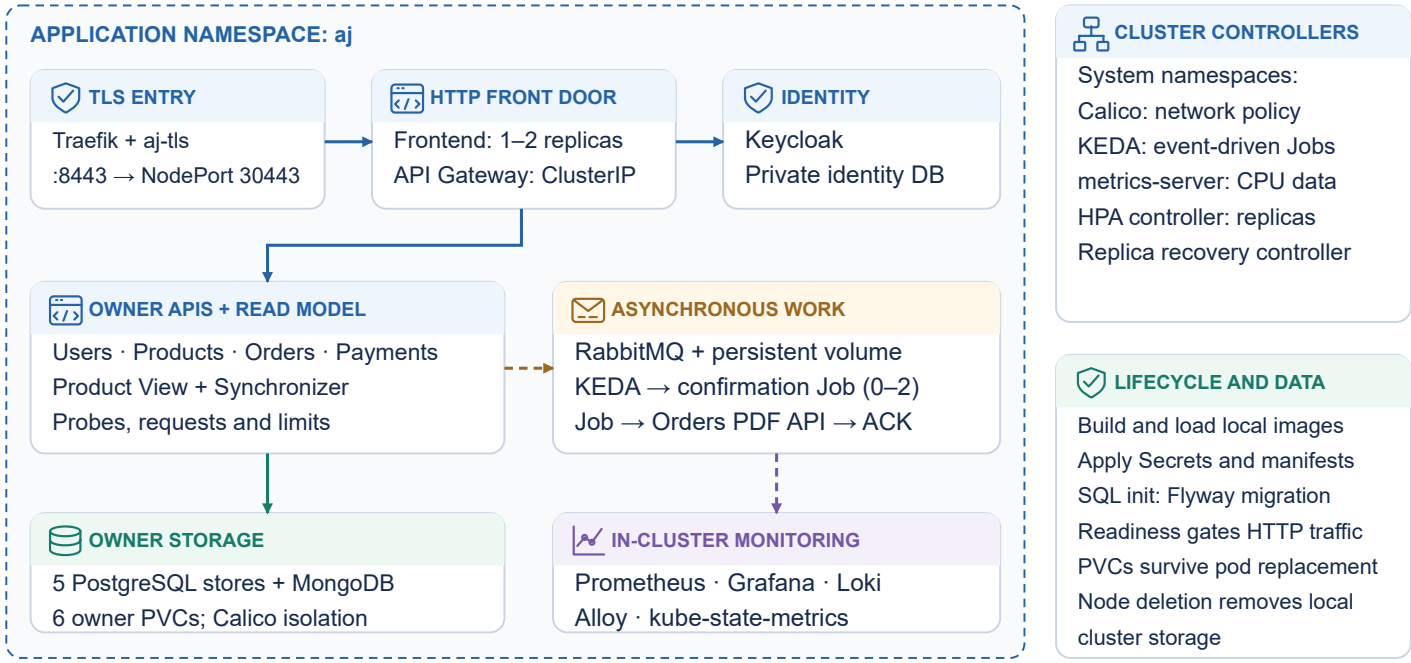
Cross-cutting checks: access control, duplicate delivery, dependency outage and retained evidence.

Kubernetes deployment

- Active demo: TLS ingress, owner PVCs, migration gates, HPA and KEDA Jobs.
- Standby: Headlamp reads resources while Compose serves the application.
- Single-node simulation is not host-failure high availability.

Local Kubernetes deployment

kind-aj-exam | one Docker-hosted node | namespace aj | separate data from Compose



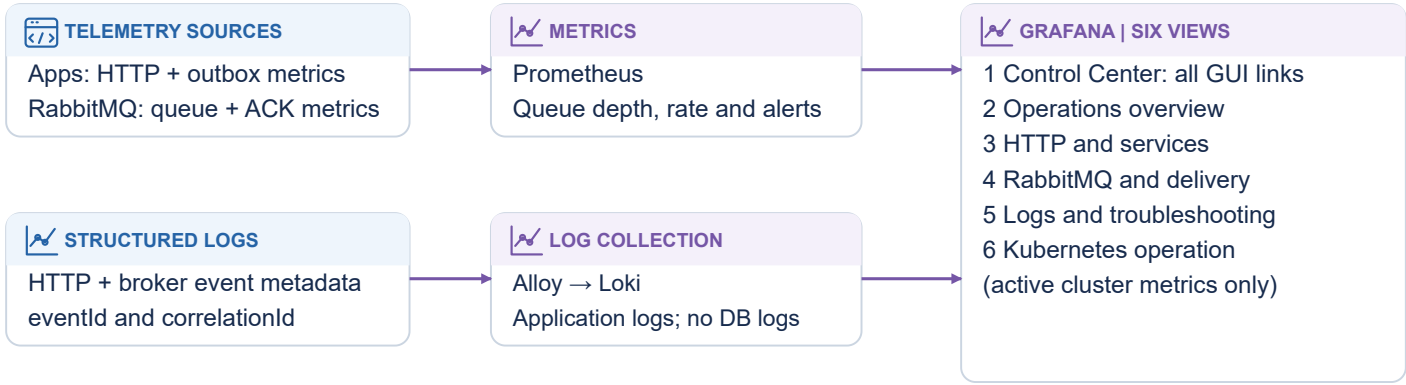
Solid: HTTP / owner storage. Dashed: asynchronous delivery or telemetry. Single-node operation is not HA.

Monitoring and GUI portal

- Grafana Control Center links SQL/Mongo/Kubernetes and all operations tools.
- Six dashboards; metrics + logs + RabbitMQ delivery status.
- Local browser checks passed all database GUIs and read-only Headlamp.

Observability and message visibility

Six Grafana dashboards: one GUI portal and five operational views



Headlamp reads cluster resources in standby; active in-cluster monitoring supplies Kubernetes metrics.

Deployment plan and delivery

- AKS cloud plan includes identity, AI, stores, broker, jobs and observability.
- Installation and verification instructions are included in the report.
- Remaining external inputs: owned OAuth app, real login and authentic contributions.
- Upload PowerPoint, report and repository link to itsLearning.